2021–2030 targets are to reach 90% fewer people requiring interventions against NTDs; ensure 75% fewer NTD-related disability adjusted life years (DALYs); 100 countries achieve elimination of at least 1NTD; and the eradication of dracunculiasis and yaws by 2030. We use Madagascar as a case study and apply a Systems Thinking approach to identify and understand the critical levers that most significantly affect, support or hinder actions towards Madagascar's NTD control and elimination goals. We focus on the context in which interventions for three NTDS - soil transmitted helminths, schistosomiasis and lymphatic filiariasis- are delivered, the role of Madagascar's NTD policy, and their NTD medicine supply chain. We examine changes in NTD outcomes based on these three main levers over a five year period using publicly available data including WHO's ESPEN portal. Findings indicate that systems thinking models can be used to identify critical lever points in health systems and inform prioritization of responses that favor desired outcomes for a country to reach its NTD goals. [1] The World Health Organization, WHO

7677

MODELLING THE EFFECT OF SEASONAL MALARIA CHEMOPREVENTION ON THE TRANSMISSION DYNAMICS OF MALARIA IN ZAMFARA STATE, NORTHWEST NIGERIA

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Seasonal malaria chemoprevention (SMC) is a highly effective intervention to prevent malaria infections in areas where the malaria burden is high, and transmission is seasonal. The study aimed to investigate the effect of seasonal malaria chemoprevention on the transmission dynamics of malaria in Zamfara state using a mathematical model. Zamfara state, in northwest Nigeria, commenced the implementation of SMC in 2016. Descriptive analysis of uncomplicated malaria cases among under-five children for 2014-2023 from the District Health Information System-2, who were diagnosed by rapid diagnostic tests or microscopy, was done. A deterministic compartmental SVEAIR for human and SEI model for mosquito populations which incorporated SMC, was formulated. Time series analysis was done. Monthly forecasts for 2024-2025 were made against the backdrop of 'business-as-usual' context. Parameter estimation was conducted using the least squares and maximum likelihood fitting techniques. Scenario analysis was done with varying SMC effectiveness levels of 25%, 45%, 65%, 90% and 100%. Between 2014-2021, there was a general upward trend of malaria cases. From 2022, there was a decline in the reported malaria cases, with slight fluctuations. This decrease is forcasted to be continued in the future (2024-2025). However, there is a possibility of increase in malaria cases if SMC effectiveness decreases or due to other factors, such as, non-adherence and drug resistance. With increasing SMC effectiveness, the malaria cases averted increased. This translates to a reduction in the number of exposed children who progress to infected status. SMC is likely to remain effective for some more years but it may take more time to reach the pre-elimination phase. There is need to sustain SMC deployment in the state in the coming years. The estimated parameters may be used for future studies in the state.

7678

INNOVATIONS IN MALARIA CAMPAIGNS IN MOZAMBIQUE: FROM DIGITALIZATION TO EVALUATION

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Campaigns have long been used by health programs to deliver services at scale. As health campaigns have grown in number, scope, and cost, the resource-constrained global health community has sought innovations to improve efficiency while maintaining effectiveness. In Mozambique this has translated to the digitalization of its insecticide treated net (ITN) campaign in 2022, followed by the seasonal malaria chemoprevention and indoor residual spraying campaigns in 2024. By instituting digitalization, Mozambique aimed to improve campaign coverage, data quality and efficiency. As workers captured data into mobile devices in the field, campaign progress was monitored in real-time, enabling fast decision-making in dedicated forums. Digital records were analyzed to assess campaign results and data quality, and coverages were validated in post-campaign surveys. In addition, 138 qualitative interviews to campaign stakeholders collected their perceptions of the new digital tools, process and their utility. Overall, data accessibility and quality improved substantially over paper-based campaigns, with over 80% of distributed commodities geolocated, and 95% of records becoming available for monitoring within 24h of collection. Data review forums were crucial for campaign management, with over 10% of the population coverage attained in the ITN campaign attributable to them. Coverage assessed in post-distribution surveys was high at 98.1-100% in target districts for ITNs. Across user interviews, 88.6% of campaign workers reported having a positive experience using the digital tools, with a further 97.1% finding them easy to use, despite historical reliance on paper systems. The experience of Mozambique has shown that digitalization has led to improved campaign performance while recording high acceptance across campaign workers. The digitalization of campaigns can strengthen their impact by enabling real-time decision-making, and through improvements in monitoring, data quality and targeting. Moreover, collected digital geolocation data is a useful asset during the planning of upcoming campaigns in targeted geographies.

7679

FREQUENCY OF HOUSEHOLD VISITS IN DEMOGRAPHIC SURVEILLANCE SYSTEM IN BANGLADESH AFFECTS ESTIMATES OF PERINATAL MORTALITY

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A demographic surveillance system has been established in September 2017 in Baliakandi sub-district of Rajbari district in Bangladesh, as an essential component of the Child Health and Mortality Prevention Surveillance (CHAMPS) project. It covers 57,259 households with a population of 237,170 in 2023, providing data on household location, member characteristics socio-economic status, women's reproductive history and demographic events. Initially, from Sep-2017 data was collected every 4 months intervals, from Oct-2018 it was changed into 2 monthly visits. After 21 months, when PSS full protocol was established and implemented data collection intervals increased to 3 months from Jul-2020. We investigated how household visits at different intervals impact the detection of birth and death events. We compared births and child deaths from Oct-17 to Sep-18 (visits every 4 months), Oct-18 to Sep-19 (visits every 2 months), and Oct-21 to Sep-22 (visits every 3 months); the years between 2019-2021 were excluded due to COVID-19. During the 4-monthly round, 4,698 new pregnancies and 4,737 deliveries were